Syllabus Year: 2024-2025

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - I

Course Code: BCA101-1 **Max Marks: 100**

Course Name: FOUNDATIONAL MATHEMATICS **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Reformulate statements from common language to formal logic using the rules of propositional and predicate calculus and assess the validity of arguments.

CO2: formulate and interpret statements presented and determine their validity by applying the rules and methods of propositional logic.

CO3: construct elementary proofs using ordinary and strong induction in the context of studying the properties of recursion.

CO4: apply basic counting principles including the pigeonhole principle and rules for counting permutations and combinations.

CO5: apply the logical structure of proofs and work symbolically with connectives and quantifiers to produce logically valid, correct and clear arguments

Section A **3*8=24**

1)	Analyze and explain why the statement 'All even integers can be expressed as 2n where n is an integer' is a valid argument in predicate calculus.	**3**	(CO1, L4)
2) a)	Examine the logical structure of the argument: 'If a number is divisible by 4, then it is even.' Construct a truth table to analyze whether this statement is a tautology.	**3**	(CO5, L4)
	OR		
b)	Analyze and differentiate between a symmetric relation and an antisymmetric relation using examples on finite sets.	**3**	(CO5, L4)
3)	Demonstrate that for any real number x, if x^2 is even, then x must also be even.	**3**	(CO5, L3)
4)	Discuss why the relation R = {(1, 2), (2, 1), (2, 3), (3, 2)} on the set A = {1, 2, 3} is symmetric but not transitive. Provide a one-line explanation for each property.	**3**	(CO3, L2)
5) a)	Explain why the statement 'All prime numbers greater than 2 are odd' can be reformulated in propositional logic using a truth table.	**3**	(CO1, L2)
	OR		
b)	Explain why the statement 'All perfect squares greater than 10 are odd' is logically valid or invalid using propositional calculus.	**3**	(CO1, L2)
6)	Determine whether the statement 'For all odd positive integers n, n^2 + 1 is even' is valid or not. Provide a proof for your answer.	**3**	(CO1, L3)

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7) a)	Analyze why the statement 'All odd positive integers greater than 1 can be expressed as the sum of a smaller odd integer and an even number' is true using elementary proof by induction.	**3**	(CO3, L4)
	OR		
b)	Examine why the statement 'All positive integers that are not divisible by 4 differ in parity from their immediate predecessor' is true using induction. Provide a base case and an inductive step to prove this claim.	**3**	(CO3, L4)
8)	Analyze and differentiate between the statements 'For all real numbers x, if x^2 - 3x + 2 > 0' and 'There exists a real number x such that x^2 - 3x + 2 = 0', providing proofs for each.	**3**	(CO5, L4)
	Section B		**8*7=56**
9)	Let = {1, 2, 3, 4}. Define a relation on as follows: = {(1,1), (2,2), (3,3), (4,4), (1,2), (2,1), (2,3), (3,2)}. Determine the following properties of and justify your answers using logical reasoning and formal logic rules (propositional or predicate calculus): (i) Is reflexive? Explain. (ii) Is symmetric? Explain. (iii) Is transitive? Explain. (iv) Determine if there exists an element in that is related to itself and another distinct element, i.e., find a pair (,) such that ≠ .	**8**	(CO1, L3)
10) a)	Determine how many positive integers less than or equal to 50 satisfy at least one of the following conditions: (i) are divisible by 3; (ii) are divisible by 4; and (iii) are not divisible by both 2 and 3. Construct a Venn diagram to illustrate your solution.	**8**	(CO4, L3)
	OR		
b)	Determine the number of onto functions from a set A = {1, 2, 3, 4} to a set B = {a, b}. Also, demonstrate that your answer satisfies the pigeonhole principle by showing why it is not possible for all elements in B to have more than one preimage under any function from A to B. (i) Compute the total number of onto functions. (ii) Explain how the pigeonhole principle is relevant to this problem.	**8**	(CO4, L3)
11)	A class contains 10 boys and 20 girls. (i) In how many ways can a committee of 5 members be formed such that it includes at least one boy? (ii) How many different ways are there to select 4 letters from the word 'MATHEMATICS' without repetition, considering the repeated letters? Explain your answer.	**8**	(CO4, L2)
12) a)	Construct elementary proofs using ordinary and strong induction to solve the following problems related to recursion and relations in sets.	**8**	(CO3, L3)
	OR		
b)	Construct elementary proofs using ordinary induction to show that for all positive integers n, the sum of the first n odd numbers is equal to n². That is, prove Σ(2−1) = ² for =1 to .	**8**	(CO3, L3)
13)	Construct elementary proofs using ordinary induction to show that the sequence defined recursively by a_1 = 2 and a_{n+1} = 3a_n + 4 for all positive integers n, satisfies the explicit formula a_n = \frac{3^n + 1}{2}. (i) Prove the base case for n=1. (ii) Assume the statement is true for some arbitrary positive integer k, i.e., a_k = \frac{3^k + 1}{2}. Use this assumption to prove that a_{k+1} = \frac{3^{k+1} + 1}{2}. (iii) Discuss how strong induction could be used as an alternative method for proving the same result.	**8**	(CO3, L2)

14) a)	A library has 10 different books on mathematics, including a special edition of 'The Elements' by Euclid. A student wishes to borrow three books for their studies. (i) In how many ways can the student select the books if they must include the special edition? (ii) If the student decides to read exactly two fiction and one non-fiction book, and there are 3 fiction and 7 non-fiction books available, in how many different ways can the student choose these three books?	**8**	(CO4, L2)
	OR		
b)	In a mathematics competition with 15 participants, each participant solves at least one problem out of four given problems. Using the pigeonhole principle and basic counting principles, (i) prove that there must be at least two participants who have solved the same set of problems. (ii) determine the minimum number of participants required to ensure that at least three participants solve exactly the same set of problems.	**8**	(CO4, L2)
15)	Analyze why the statement 'In a class of 10 students, at least two have their birthdays in the same month' must always be true, using principles from counting and permutations. Next, consider a scenario where 6 letters are to be distributed into 3 distinct mailboxes: (i) Determine the total number of ways this can be done without any restrictions. (ii) Calculate how many ways this can be done if no mailbox is left empty.	**8**	(CO4, L4)
	Section C		**10*2=20**
16)	Analyze why the statement 'All positive integers that are multiples of 3 can be expressed as the sum of two or more consecutive positive integers' is true by constructing a truth table for the propositional logic involved in the proof. Additionally, examine and differentiate between two methods to prove this statement: one using algebraic manipulation and another using logical reasoning.	**10**	(CO2, L4)
17) a)	Demonstrate through propositional logic that the statement 'For all positive integers n, if n is odd then 2n + 1 is also odd' holds true by constructing a logical argument. Additionally, determine how many such statements are true for the first 10 positive integers.	**10**	(CO2, L3)
	OR		
b)	Consider the set A = {1, 2, 3, 4}. Let R be a relation on A defined by R = {(1,1), (2,2), (3,3), (4,4), (1,3), (3,1)}. Determine whether R is an equivalence relation. If it is, provide the equivalence classes and show that they are finite. Then, construct a directed graph to represent this relation and use it to verify the properties of R.	**10**	(CO2, L3)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create

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